The psql Cheatsheet

A friendly, visual guide to PostgreSQL's interactive terminal

PostgreSQL 16+ | Beginner → Pro | Backslash · SQL · Tricks | 2026 Edition

1 Get Connected

⚡ Anatomy of a connection

USER	HOST	PORT	DATABASE
`-U alice`	`-h db.io`	`-p 5432`	`-d shop`

```
# classic flag form
psql -h db.io -p 5432 -U alice -d shop

# URI form (great for scripts & copy-paste)
psql "postgresql://alice:secret@db.io:5432/shop?sslmode=require"

# env-var form (no secrets in shell history)
PGPASSWORD=secret psql -U alice -d shop
```

Pro tip: store credentials in `~/.pgpass` with `chmod 600` and skip the password prompt forever.

↻ Switch / reconnect inside psql

`\c db`	connect to another database
`\c db user`	connect as another user
`\conninfo`	show current connection details
`\password`	set password securely (no echo)
`\q` · `Ctrl` + `D`	quit psql

? Where am I?

`SELECT current_database();`	db name
`SELECT current_user;`	login role
`SELECT current_schema();`	active schema
`SELECT version();`	server version
`SHOW search_path;`	name resolution order

2 Explore the Schema

i Anatomy of a backslash command

```
\d+ public.users
```

backslash starts a meta-command · verb · what to do (d = describe) · modifier · + verbose, S system · argument · object pattern (* ?)

Describe objects

`\d`	list tables, views, sequences
`\d name`	describe one object
`\d+ name`	verbose: storage, stats
`\dt`	tables only
`\dv`	views
`\di`	indexes
`\ds`	sequences
`\dm`	materialized views
`\df`	functions
`\dn`	schemas (namespaces)
`\du`	users / roles
`\dx`	installed extensions
`\dp`	privileges per object

Patterns: `\dt public.*` · append S for system (`\dtS`).

See what's inside

```
\dt+                    -- tables w/ size
\d users                -- columns + indexes + FKs
\d+ users               -- + storage, comments, stats
\sf get_user(int)       -- show function source
\sv active_users        -- show view definition
\ef get_user(int)       -- edit function in $EDITOR
```

Fun fact: `\d` is short for *describe*. Add letters to narrow scope, modifiers to widen it.

3 Run Queries Like a Pro

The query buffer

`;`	execute current statement
`\g [file]`	re-run last query (or save)
`\gx`	run + force expanded
`\gexec`	run, then exec each row as SQL
`\watch n`	re-run every n seconds
`\e`	edit last query in $EDITOR
`\p · \r · \w`	print · reset · write buffer
`\i file.sql`	run commands from a file
`\ir file.sql`	include relative to script

Magic of `\gexec`

```
-- Drop every table starting with "tmp_"
SELECT 'DROP TABLE ' || tablename || ';'
FROM pg_tables
WHERE tablename LIKE 'tmp\_%'
\gexec
```

Heads up: `\gexec` runs whatever your query returned. Test with SELECT first, then swap in the action.

Keyboard shortcuts (readline)

`Ctrl` + `R`	reverse-search history	`Ctrl` + `A`	jump to start of line	`Ctrl` + `L`	clear screen
`↑` `↓`	walk through history	`Ctrl` + `E`	jump to end of line	`Ctrl` + `C`	cancel current input
`Tab`	autocomplete tables, cols, kw	`Ctrl` + `U`	clear line before cursor	`Ctrl` + `D`	quit on empty line

4 Make Output Beautiful

Output formats at a glance

Aligned (default)

```
 id | name  | age
----+-------+-----
  1 | alice |  30
  2 | bob   |  27
```

Expanded · `\x`

```
-[ RECORD 1 ]-
id   | 1
name | alice
age  | 30
```

CSV · `--csv`

```
id,name,age
1,alice,30
2,bob,27
```

Formatting toggles

`\x [on|off|auto]`	expanded display
`\pset format fmt`	aligned · csv · html · latex
`\pset border 2`	draw full borders
`\pset null 'ø'`	placeholder for NULLs
`\pset pager off`	disable pager (good in scripts)
`\pset linestyle unicode`	pretty box-drawing borders
`\pset numericlocale on`	thousand separators
`\timing on`	show duration of every query

Send results elsewhere

`\o file.txt`	redirect output to file
`\o | less`	pipe to a shell command
`\o`	stop redirecting (back to term)
`\g file.csv`	save just that result
`\H`	toggle HTML output mode
`\copy ... TO ...`	client-side CSV export (§6)

Tip: `\timing on` + `\watch 2` = a free dashboard.

5 Variables & Scripting

Set & use variables

```
\set uid 42
\set tbl 'users'

-- numeric / identifier substitution
SELECT * FROM :tbl WHERE id = :uid;

-- quote a value safely
SELECT * FROM users WHERE name = :'name';

-- quote an identifier safely
SELECT * FROM :"tbl";
```

`\set`	list all variables
`\unset name`	remove variable
`\echo :var`	print value (debugging)

Conditional & safer scripts

```
\set ON_ERROR_STOP on   -- abort on first error
\set AUTOCOMMIT off

\if :{?some_var}
  SELECT 'var is set';
\else
  \echo 'please pass -v some_var=...'
\endif
```

Recipe: launch with `psql -v ON_ERROR_STOP=1 -1 -f migrate.sql` to wrap a whole script in a single transaction.

6 Import & Export Data

CSV in & out with \copy

```
-- Export a table
\copy users TO 'users.csv' CSV HEADER

-- Export a query result
\copy (SELECT id, email FROM users
       WHERE active) TO 'active.csv'
      CSV HEADER

-- Import (skips header)
\copy users FROM 'users.csv'
      CSV HEADER NULL ''
```

\copy vs COPY: `\copy` reads files on *your* machine. `COPY` uses *server* files and needs superuser perms.

Backup & restore (shell)

```
# dump a single database
pg_dump -Fc shop > shop.dump

# restore
pg_restore -d shop_new shop.dump

# dump + restore over a pipe
pg_dump shop | psql shop_clone

# schema only / data only
pg_dump --schema-only shop > ddl.sql
pg_dump --data-only   shop > rows.sql
```

7 Diagnose & Tune

Health checks & lookups

```
-- who is connected right now?
SELECT pid, usename, datname, state, query
FROM pg_stat_activity
WHERE state <> 'idle';

-- biggest tables
SELECT relname,
       pg_size_pretty(pg_total_relation_size(oid))
FROM pg_class
WHERE relkind = 'r'
ORDER BY pg_total_relation_size(oid) DESC
LIMIT 10;
```

```
-- ask the planner why
EXPLAIN (ANALYZE, BUFFERS)
SELECT * FROM users WHERE email = 'a@b.io';

-- kill a runaway query
SELECT pg_cancel_backend(12345);    -- polite
SELECT pg_terminate_backend(12345); -- forceful

-- locks & blockers
SELECT * FROM pg_locks WHERE NOT granted;
```

Built-in help

`\?`	all psql meta-commands
`\? variables`	list of psql variables
`\h`	all SQL command names
`\h SELECT`	syntax for any keyword
`\copyright`	distribution terms

Fun & useful tricks

- Run `\watch 1` after a slow query → live refresh.
- Make a `~/.psqlrc` with `\timing on` and a custom `PROMPT1`.
- Press `Tab` `Tab` on an empty line for everything you can complete.
- `\! ls` runs a shell command without leaving psql.
- `\dconfig` shows server settings; `\dconfig+` for sources.

Built with ♥ for PostgreSQL · keep this open in a tab next to your terminal · \? is your best friend